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Interbank Liquidity Crunch and the Firm Credit Crunch: Evidence from the
2007–2009 Crisis

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1 Big picture

- **Question.** Does an interbank liquidity shock reduce bank credit supply to firms, and is the effect especially binding for smaller entrepreneurial firms?
- **Setting.** The paper studies Portugal during the 2007–2009 crisis. The key shock is the unexpected freeze of the European interbank market beginning in August 2007.
- **Core idea.** Banks that relied more on interbank borrowing before the crisis were more exposed to the sudden dry-up in wholesale liquidity. The paper asks whether those banks cut lending more after the shock.
- **Data advantage.** The authors use exhaustive Portuguese credit-register data, matched to bank balance sheets and firm balance-sheet information. This lets them observe lending at the firm-bank level and compare different banks lending to the same firm.
- **Identification.** The baseline design uses firm fixed effects. For the same borrower, the paper compares the change in credit from a high-interbank-borrowing bank with the change in credit from a low-interbank-borrowing bank.
- **Main credit-supply result.** Banks with higher precrisis interbank borrowing reduce firm credit more during the crisis. A 10 percentage point increase in precrisis interbank borrowing implies about a 3.7–4 percent additional reduction in firm credit.
- **Entrepreneurial-firm result.** The contraction is concentrated among smaller, younger firms and firms with weaker preexisting banking relationships.
- **Extensive-margin result.** Liquidity-exposed banks are less likely to start new lending relationships with small firms, and stronger existing relationships reduce the likelihood that loans are terminated.
- **Bank balance-sheet result.** The negative effect of bank illiquidity is stronger for less solvent banks, especially banks with higher nonperforming loans.
- **Central-bank liquidity result.** Central bank borrowing does not clearly offset the credit crunch for highly exposed banks. The evidence is consistent with liquidity hoarding: exposed banks increase liquid assets rather than expanding credit.
- **Binding-constraint result.** Firms cannot fully substitute lost credit from affected banks with credit from less affected banks, trade credit, or other debt. The credit crunch is therefore binding for firms.
- **Economic takeaway.** Interbank-market stress can transmit to the real economy through bank lending, and the burden falls disproportionately on small entrepreneurial firms that cannot easily replace bank finance.

2 Incremental contribution of the paper

- **From financial crisis to firm-level credit supply.** The paper links an interbank liquidity shock directly to firm credit availability, using micro data rather than aggregate lending trends.

- **Clean use of loan-level firm fixed effects.** Building on the Khwaja-Mian credit-supply approach, the paper compares lending by different banks to the same firm, which absorbs firm-level loan demand, risk, and quality.
- **A crisis-specific liquidity shock.** The August 2007 interbank freeze was unexpected and external to Portuguese banks. Portuguese banks were not heavily exposed to U.S. subprime, dollar funding, or off-balance-sheet vehicles, making Portugal a useful setting for isolating interbank liquidity dependence.
- **Entrepreneurial finance contribution.** The paper shows that the credit-supply shock is not uniform. It is strongest for smaller and younger firms, the firms most likely to rely on bank relationships and least able to access market finance.
- **Extensive-margin contribution.** Beyond intensive-margin credit changes, the paper studies new lending relationships and termination of existing relationships. This matters because small firms often need new bank relationships to grow.
- **Solvency-liquidity interaction.** The analysis shows that liquidity shocks are amplified by weak bank fundamentals. Banks with higher nonperforming loans cut credit more when exposed to interbank funding stress.
- **Policy contribution.** The paper studies whether central bank liquidity provision alleviates the credit crunch. The evidence suggests that liquidity injections can be absorbed as bank liquidity buffers rather than being passed through to firms.
- **Binding-credit-crunch contribution.** By moving from firm-bank regressions to firm-level total credit and total debt, the paper shows that affected firms cannot fully offset the loss of bank credit.
- **A complete transmission chain.** The paper traces the chain from interbank liquidity dry-up to bank credit-supply contraction, to small-firm credit constraints, to incomplete substitution through other financing sources.

3 Data and measurement

3.1 Portuguese setting and interbank shock

- The crisis began in Europe in August 2007, when the interbank market experienced severe tensions and interbank spreads rose sharply.
- The European Central Bank injected large amounts of liquidity after the interbank market froze.
- The shock is plausibly external to Portuguese banks because they had little direct exposure to U.S. subprime assets, dollar funding stress, or off-balance-sheet vehicles funded by asset-backed commercial paper.
- Portugal also did not experience a large precrisis real estate boom comparable to Ireland, Spain, the United Kingdom, or the United States.
- The key cross-sectional exposure variable is the bank's **interbank borrowing ratio** before the crisis, measured as interbank borrowing divided by total assets in 2007:Q2.

Interpretation: the paper treats precrisis interbank dependence as a measure of how severely each bank was hit by the unexpected liquidity shock.

3.2 Credit-register data

- The main data source is the Portuguese Central Credit Register, maintained by the Bank of Portugal.
- The register covers all commercial and industrial loans to nonfinancial publicly limited and limited-liability firms.
- The reporting threshold is only 50 euros, which makes the data especially useful for studying small firms.
- The authors aggregate all loans between each firm-bank pair in 2007:Q2 and 2009:Q2.
- The main dependent variable is the change in the log of committed credit from bank b to firm i between 2009:Q2 and 2007:Q2.

Interpretation: the loan-level panel allows the authors to study changes in credit supplied by each bank to each firm, rather than only aggregate credit volumes.

3.3 Bank variables

The paper matches the credit register to complete bank balance-sheet data. Main bank variables include:

- interbank borrowing ratio;
- foreign and domestic components of interbank borrowing;
- bank size;
- liquid assets;
- return on assets;
- nonperforming loans;
- bank capital;
- central bank borrowing from the ECB;
- changes in bank total assets and liquid assets.

Interpretation: these variables allow the paper to distinguish liquidity exposure from bank size, profitability, liquidity buffers, capital, and loan-quality problems.

3.4 Firm variables

The firm-level balance-sheet information includes:

- total assets;

- number of employees;
- firm age;
- return on assets;
- interest coverage ratio;
- total debt;
- industry and location;
- the amount of precrisis credit between each firm-bank pair.

Interpretation: these variables are used to ask which firms bear the credit contraction and whether affected firms can substitute toward other credit sources.

3.5 Sample construction

- The full firm-bank sample contains 293,896 firm-bank pairs.
- The firm-fixed-effect identification relies on firms with at least two banking relationships; this leaves 220,406 firm-bank pairs in the main sample.
- The bank sample focuses on the 84 largest banks, which account for roughly 90 percent of total lending in the Portuguese banking system.
- The average firm has about 3.19 banking relationships before the crisis.

Interpretation: multiple-bank firms are crucial because firm fixed effects compare different lenders to the same borrower.

4 Models and empirical specifications

4.1 Baseline firm-bank credit-supply regression

The baseline specification is:

$$\Delta \log Credit_{bi} = \alpha_i + \beta InterbankBorrowing_{b,2007:Q2} + \Gamma' BankControls_{b,2007:Q2} + \varepsilon_{bi}.$$

- $\Delta \log Credit_{bi}$ is the change in committed credit from bank b to firm i between 2009:Q2 and 2007:Q2.
- α_i is a firm fixed effect.
- $InterbankBorrowing_{b,2007:Q2}$ measures the bank's precrisis interbank borrowing ratio.
- Bank controls include bank size, liquid assets, return on assets, nonperforming loans, and capital.

The coefficient of interest is β . A negative β means that banks more dependent on interbank funding before the crisis cut lending more during the crisis.

Interpretation: firm fixed effects absorb firm loan demand, risk, quality, and investment opportunities. Identification comes from within-firm differences across lenders.

4.2 Firm heterogeneity regressions

To ask which firms are most affected, the paper adds interactions between bank interbank borrowing and firm characteristics:

$$\Delta \log Credit_{bi} = \alpha_i + \beta InterbankBorrowing_b + \theta(InterbankBorrowing_b \times FirmCharacteristic_i) + \Gamma' BankControls$$

Firm characteristics include size, age, employment, interest coverage, ROA, and precrisis firm-bank credit.

Interpretation: a positive interaction with firm size or age means that the negative effect of bank liquidity exposure is weaker for larger or older firms.

4.3 Small-versus-large firm threshold regressions

The paper also estimates threshold regressions:

$$\Delta \log Credit_{bi} = \alpha_i + \beta_S(InterbankBorrowing_b \times 1\{Small_i\}) + \beta_L(InterbankBorrowing_b \times 1\{Large_i\}) + \Gamma' BankControls$$

The thresholds are based on total assets and number of employees. The key test is whether β_S is negative and significant while β_L is close to zero.

Interpretation: this directly tests whether the bank liquidity shock affects entrepreneurial firms but not large firms.

4.4 Extensive-margin regressions

The paper studies two extensive margins:

- **New lending relationship:** equals one if a firm has a loan in 2009:Q2 from a bank with which it had no loan in 2007:Q2.
- **Termination of existing relationship:** equals one if a firm had a loan from a bank in 2007:Q2 and has no loan from that bank in 2009:Q2.

The regressions interact interbank borrowing with firm-size indicators and relationship-strength measures.

Interpretation: intensive-margin credit contractions are only part of the credit crunch. Small firms may also lose access to new banking relationships.

4.5 Bank solvency, central bank borrowing, and portfolio rebalancing

The paper tests whether the effect of interbank exposure depends on bank fundamentals:

$$\Delta \log Credit_{bi} = \alpha_i + \beta InterbankBorrowing_b + \theta(InterbankBorrowing_b \times BankCharacteristic_b) + \Gamma' BankControls$$

Bank characteristics include size, liquid assets, ROA, nonperforming loans, capital, and central bank borrowing.

The bank-level portfolio regressions study:

$$\Delta TotalAssets_b \quad \text{and} \quad \Delta LiquidAssets_b$$

as functions of precrisis interbank borrowing.

Interpretation: if exposed banks receive or conserve liquidity but still cut lending, liquidity provision may produce hoarding rather than credit supply to firms.

4.6 Firm-level substitution regressions

To test whether the credit crunch is binding, the paper constructs a firm-level exposure measure:

$$AverageInterbankBorrowing_i = \sum_b w_{bi,2007:Q2} InterbankBorrowing_b,$$

where $w_{bi,2007:Q2}$ is the bank's share in firm i 's precrisis credit.

The firm-level regressions are:

$$\Delta \log TotalBankDebt_i = \alpha + \beta AverageInterbankBorrowing_i + X_i' \Gamma + FE + \varepsilon_i,$$

and analogous specifications for total firm debt.

Interpretation: if firms fully substitute away from affected banks, average bank exposure should not reduce total bank debt or total debt. A negative coefficient means the credit supply shock is binding.

5 Identification and empirical design

5.1 Identifying credit supply rather than credit demand

The central empirical challenge is separating supply from demand. During a crisis, firms may demand less credit, become riskier, or reduce investment. A simple bank-level or firm-level regression could confuse these demand effects with bank credit-supply effects.

The firm fixed effects solve this by comparing different banks lending to the same firm. If the same firm receives a larger credit cut from the bank with higher precrisis interbank borrowing, the difference is hard to explain by firm demand.

Interpretation: the paper identifies relative credit supply across banks for the same borrower.

5.2 Why precrisis interbank borrowing is useful

Precrisis interbank borrowing measures exposure to the liquidity shock before the shock occurs. The paper emphasizes that this variable is not strongly correlated with precrisis bank profitability, risk, or capital. This supports the interpretation that it captures liquidity exposure rather than simply weak banks choosing risky lending.

Interpretation: the research design uses predetermined bank exposure to an unexpected funding shock.

5.3 Why Portugal is a useful setting

Portugal is useful because:

- banks relied on foreign interbank funding, so the European interbank freeze mattered;
- banks were not at the center of U.S. subprime losses;
- regulation did not encourage off-balance-sheet vehicles funded by ABCP;
- Portugal did not have a major precrisis real estate boom;
- the credit register observes essentially all business loans, including very small ones.

Interpretation: the setting helps isolate the interbank liquidity channel from other crisis mechanisms.

5.4 Testing whether the credit contraction is binding

The firm-bank regressions identify a bank credit-supply reduction, but firms might offset that reduction by borrowing from other banks or using other debt. The paper therefore estimates firm-level regressions using total bank debt and total firm debt.

Interpretation: this step converts a bank-level supply shock into a firm-level financing constraint.

5.5 Limitations

- The firm fixed effects identify relative loan-supply changes across banks, not every aggregate change in credit supply.
- The analysis covers Portuguese firms and banks, so external validity depends on whether other banking systems have similar dependence on interbank funding and relationship lending.
- The event window captures the 2007–2009 crisis, during which many macro shocks occurred. The firm fixed effects and predetermined exposure help but cannot eliminate all aggregate-credit-market concerns.
- The paper measures credit availability through committed credit volumes; drawn credit and unused lines may behave differently.
- Central-bank liquidity regressions are informative but not a randomized policy experiment. Banks that borrow from the ECB may differ in unobserved ways.

Interpretation: the paper is strongest on the relative credit-supply channel and less definitive on the welfare effect of all crisis policies.

6 Tables and figures

6.1 Figures 1 and 2: aggregate credit growth and interbank borrowing

Read:

- Figure 1 shows growth in total credit volume to nonfinancial firms. Credit growth rises before the crisis, peaks around the crisis period, and then falls sharply through 2008 and 2009.
- Figure 2 shows interbank borrowing as a fraction of total assets. Interbank borrowing is relatively stable before the crisis and then declines as the interbank market stress intensifies.
- The timing is important: interbank borrowing begins falling before the sharp decline in firm credit growth.
- The figures motivate the empirical question but do not identify supply versus demand. The loan-level regressions do that.

Economic message: The raw time series suggest a funding-to-lending transmission channel: the interbank liquidity crunch precedes and accompanies the firm credit crunch.

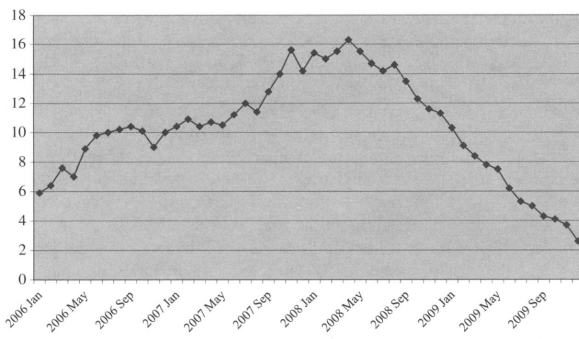


Figure 1
Growth in total credit volume for nonfinancial firms

(a) Figure 1: growth in total credit volume for nonfinancial firms

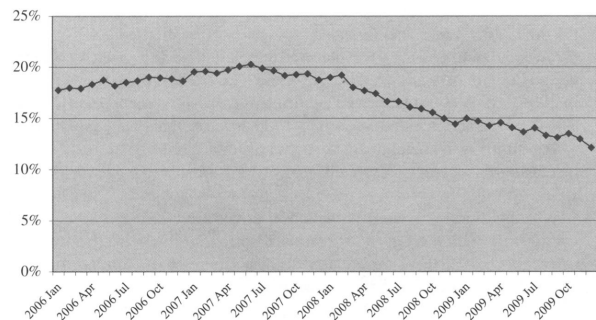


Figure 2
Interbank borrowing (as a fraction of total assets)

(b) Figure 2: interbank borrowing as a fraction of total assets

6.2 Tables 1 and 2: summary statistics and baseline credit-supply reduction

Read:

- Table 1 reports an average of 3.19 banking relationships per firm and an average credit change of -0.16 at the firm-bank level.
- Average interbank borrowing is 0.25 of bank assets, showing substantial exposure to interbank funding before the crisis.
- The summary statistics also show substantial variation in bank and firm characteristics, which supports heterogeneous-effect tests.
- Table 2 is the baseline credit-supply table. Interbank borrowing is negative and significant across specifications.
- With firm fixed effects and bank controls, the coefficient is about -0.371 , implying that a 10 percentage point increase in interbank borrowing reduces credit by about 3.7 percent.
- Adding firm fixed effects does not eliminate the coefficient, which supports the interpretation that the effect is bank supply rather than borrower demand.

Economic message: Banks more exposed to the interbank market before the crisis cut credit more afterward, even when comparing different banks lending to the same firm.

Table 1
Summary statistics

	Mean	Median	SD
No. of banking relationships	3.19	2	2.55
Change in credit	-0.16	-0.02	1.20
Interbank borrowing	0.25	0.05	0.31
Foreign interbank borrowing	0.27	0.18	0.26
Domestic interbank borrowing	0.12	0.006	0.13
Bank size	19.90	19.54	1.95
Bank liquid assets	0.29	0.31	0.22
Bank return on assets (ROA)	0.005	0.004	0.00
Bank non-performing loans (NPL)	0.009	0.004	0.02
Bank capital	0.11	0.09	0.10
Firm size	13.11	12.97	1.64
Firm employees	1.918	1.791	1.20
Firm age	2.470	2.484	0.77

No. of banking relationships is the number of different banks that lend to a firm as of 2007:Q2, conditioned on whether the firm has at least one outstanding bank loan. *Change in credit* is the change in the (log) level of all the loans for each firm-bank pair between 2009:Q2 and 2007:Q2, where the initial interbank shock was August 2007. *Interbank borrowing* is the ratio of total interbank borrowing in 2007:Q2 for a bank; *Foreign interbank borrowing* is the part obtained abroad; and *Domestic interbank borrowing* is the part obtained from other bank in Portugal. *Size* is the log of total assets of the bank. *Liquid assets* are the ratio of cash to total assets of the bank. *Return on assets* is the profits of the bank divided by its total assets. *Nonperforming* loans is the fraction of loans that are in default as a fraction of total assets of the bank. *Capital* is the total bank capital as a fraction of total assets. Bank variables are as of 2007:Q2. *Firm size* is the log of total assets of the firm. *Firm age* is the age of the firm measured using the date of the firm's incorporation ($\ln(1+\text{age})$). *Employees* is the log of number of employees of a firm. Firm-level variables are measured as of December 2006.

(a) Table 1: summary statistics

Table 2
Credit supply reduction and total interbank borrowing

Change in the log level of firm-bank total credit volume between 2009:Q2 and 2007:Q2

	(1)	(2)	(3)	(4)	(5)	(6)
Interbank borrowing	-0.558*** (0.118)	-0.556*** (0.123)	-0.329*** (0.100)	-0.623*** (0.122)	-0.371*** (0.112)	-0.554*** (0.129)
Bank size			0.065*** (0.019)		0.070*** (0.019)	0.087 (0.024)
Bank liquid assets			0.284 (0.291)		0.337 (0.345)	0.320 (0.437)
Bank return on assets			11.087 (7.142)		11.252 (8.967)	0.643 (9.553)
Bank nonperforming assets			3.181** (1.347)		3.678*** (1.267)	3.983*** (1.346)
Bank capital			-0.472 (0.857)		-0.542 (0.948)	-0.859 (1.207)
Total firm-bank credit 2007:Q2						-0.207*** (0.017)
Adj. R^2	0.012	0.020	0.020	0.086	0.095	0.156
No. of banking relationships > 1	no	yes	yes	yes	yes	yes
Firm fixed effects	no	no	no	yes	yes	yes
Number of obs.	293,896	220,406	220,406	220,406	220,364	220,364

This table presents results of OLS regressions, where the dependent variable is the change in the (log) level of committed credit volume for each firm-bank pair between 2009:Q2 and 2007:Q2, where the initial interbank shock was August 2007. *Interbank borrowing* is the ratio of interbank borrowing to total assets in 2007:Q2 for a bank. *Size* is the log of total assets of the bank. *Liquid assets* is the ratio of short-term to total assets of the bank. *Return on assets* is the profits of the bank divided by its total assets. *Nonperforming* loans is the fraction of loans that are in default as a fraction of total assets of the bank. *Capital* is the total bank capital as a fraction of total assets. Total firm-bank credit is the log of committed credit outstanding between a firm and a bank as of 2007:Q2. Bank variables are as of 2007:Q2. Standard errors, reported in parentheses, are clustered at the bank level and are heteroscedasticity consistent. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

(b) Table 2: credit supply reduction and total interbank borrowing

6.3 Tables 3 and 4: heterogeneous firm effects

Read:

- Table 3 interacts interbank borrowing with firm characteristics while keeping firm fixed effects.
- The positive interaction with firm size means that larger firms experience a weaker negative effect from exposed banks.
- The positive interaction with firm age means that younger firms are hit harder.
- The positive interaction with firm-bank credit means that stronger precrisis relationships reduce the effect of the liquidity shock.
- The interaction with interest coverage is negative: banks cut credit less to weaker firms with lower interest coverage, consistent with possible evergreening or support of weak borrowers to avoid defaults.
- Table 4 splits firms by size thresholds. The interbank-borrowing effect is negative and significant for firms below the size thresholds, but insignificant and often close to zero for larger firms.

Economic message: The credit crunch is not uniform. It is concentrated in entrepreneurial firms: small, young firms with weaker bank relationships.

Table 3

Credit supply reduction across different firms and total interbank borrowing

Change in the log level of firm-bank total credit volume between 2009:Q2 and 2007:Q2

	(1)	(2)	(3)	(4)	(5)	(6)
Interbank borrowing	-1.521*** (0.452)	-0.563*** (0.146)	-0.499*** (0.124)	-0.365*** (0.119)	-0.365*** (0.117)	-0.435* (0.110)
Firm size * Interbank borr	0.083** (0.034)					
Firm age * Interbank borr		0.080** (0.038)				
Firm employees * Interbank borr			0.056 (0.037)			
Firm int coverage * Interbank borr				-0.075*** (0.011)		
Firm ROA * Interbank borr					-0.140 (0.118)	
Firm-bank credit * Interbank borr						0.088* (0.019)
Adj. R^2	0.095	0.095	0.095	0.095	0.095	0.095
Firm fixed effects	yes	yes	yes	yes	yes	yes
Number of obs	154,434	154,434	154,434	154,434	154,434	154,434

This table presents results of OLS regressions, where the dependent variable is the change in the (log) level of committed credit volume for each firm-bank pair between 2009:Q2 and 2007:Q2, where the initial interbank shock was August 2007. *Interbank borrowing* is the ratio of interbank borrowing to total assets in 2007:Q2 for bank. *Firm size* is the log of total assets of the firm. *Firm age* is the age of the firm measured using the date of its incorporation ($\ln(1+age)$). *Employees* is the log of number of employees of a firm. *Firm interest coverage* is defined as firm gross profit divided by interest expense on loans. *Firm ROA* is the profits of the firm divided by its total assets. *Firm-bank credit* is the log of credit outstanding between a firm and a bank as of 2007:Q2. The sample is restricted in all regression to firms that have at least two banking relationships. Regression include controls for bank characteristics (unreported), and in Column 6, we also include the bank-firm credit as of 2007:Q2 as an independent variable. Firm-level variables are measured as of December 2006. Standard errors, reported in parentheses, are clustered at the bank level and are heteroscedasticity consistent. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

(a) Table 3: credit supply reduction across different firms

Table 4

Credit supply reduction, interbank borrowing, and firm size (total assets and number of employees)

Change in the log level of firm-bank total credit volume between 2009:Q2 and 2007:Q2

	Dummy cutoff for firm total assets			Dummy cutoff for no. of firm employees		
	1,000,000 (1)	10,000,000 (2)	100,000,000 (3)	50 (4)	100 (5)	250 (6)
Interbank borr * dummy less than size/no of employees	-0.508*** (0.120)	-0.406*** (0.116)	-0.381*** (0.118)	-0.415*** (0.116)	-0.399*** (0.117)	-0.392*** (0.119)
Interbank borr * dummy more than size/no. of employees	-0.213 (0.138)	-0.048 (0.219)	0.140 (0.480)	-0.164 (0.187)	-0.144 (0.215)	0.001 (0.293)
Bank size	0.076*** (0.020)	0.076*** (0.020)	0.076*** (0.020)	0.077*** (0.020)	0.077*** (0.020)	0.077*** (0.020)
Bank liquid assets	0.361 (0.377)	0.379 (0.378)	0.390 (0.380)	0.370 (0.379)	0.373 (0.381)	0.374 (0.382)
Bank return on assets	11.970 (9.621)	12.173 (9.597)	12.341 (9.604)	12.484 (9.467)	12.475 (9.636)	12.464 (9.646)
Bank nonperforming assets	3.844*** (1.238)	3.997*** (1.290)	4.101*** (1.331)	4.086*** (1.304)	4.140*** (1.324)	4.174*** (1.337)
Bank capital	-0.545 (1.035)	-0.622 (1.035)	-0.657 (1.031)	-0.658 (1.041)	-0.675 (1.040)	-0.684 (1.039)
Adj. R^2	0.110	0.110	0.110	0.105	0.105	0.104
Firm fixed effects	yes	yes	yes	yes	yes	yes
Number of Obs.	194,862	194,862	194,862	187,205	187,205	187,205

This table presents results of OLS regressions, where the dependent variable is the change in the (log) level of committed credit volume for each firm-bank pair between 2009:Q2 and 2007:Q2. We use different firm size thresholds measured as of December 2006 to create dummy variables. For example, in Column 1, we have two dummy variables that we interact with interbank borrowing: the first one is a dummy for firms with total assets less than one million euros, and the second is a dummy for firms with total assets more than one million euros. Columns 1 to 3 use thresholds based on total assets of the firm. Columns 4 to 6 use thresholds based on the number of firm employees as of December 2006. *Interbank borrowing* is the ratio of interbank borrowing in 2007:Q2 for a bank. *Size* is the log of total assets of the bank. *Liquid assets* is the ratio of short-term to total assets of the bank. *Return on assets* is the profits of the bank divided by its total assets. *Nonperforming* loans is the fraction of loans that are in default as a fraction of total assets of the bank. *Capital* is the total bank capital as a fraction of total assets. All the bank variables are as of 2007:Q2. The sample is restricted in all regression to firms that have at least two banking relationships. Standard errors, reported in parentheses, are clustered at the bank level and are heteroscedasticity consistent. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

(b) Table 4: interbank borrowing and firm size thresholds

6.4 Tables 5 and 6: extensive margins and relationship strength

Read:

- Table 5 studies the extensive margin of new relationships and termination of existing relationships.
- High-interbank-borrowing banks are less likely to start new lending relationships with small firms.
- The new-relationship effect is not present for larger firms, suggesting that small firms lose access to new credit sources when exposed banks become illiquid.
- The termination regressions in Table 5 do not show a strong small-versus-large difference by themselves.
- Table 6 adds relationship strength. Higher interbank borrowing raises the probability of termination, but stronger bank-firm relationships reduce that probability.
- The protective role of relationship strength is similar across small and large firms.

Economic message: Relationship lending matters during liquidity crises. Small firms are disadvantaged in forming new relationships, while strong existing relationships protect borrowers from termination.

Table 5
The extensive margin, interbank borrowing, and firm size

	New lending relationships		Termination of existing lending relationship			
	Size	No. of employees	Size	No. of Employees		
	(1)	(2)	(3)	(4)	(5)	(6)
Interbank borrowing	-0.008 (0.005)			0.083 (0.057)		
Interbank borr * dummy less than size/no. of employees		-0.009* (0.005)	-0.008* (0.005)		0.066 (0.060)	0.068 (0.061)
Interbank Borr * dummy more than size/no. of employees		0.003 (0.009)	-0.004 (0.008)		0.085 (0.056)	0.081 (0.058)
Bank size	0.001 (0.0008)	0.001 (0.0008)	0.001 (0.0008)	-0.028*** (0.007)	-0.030*** (0.007)	-0.031*** (0.007)
Bank liquid assets	-0.004 (0.009)	-0.003 (0.010)	-0.002 (0.010)	-0.040 (0.114)	-0.077 (0.128)	-0.079 (0.130)
Bank return on assets	1.284** (0.579)	1.464** (0.578)	1.473** (0.580)	-0.529 (5.513)	-0.771 (5.439)	-0.867 (5.497)
Bank nonperforming assets	0.043 (0.047)	0.053 (0.049)	0.056 (0.049)	0.539 (1.334)	0.434 (1.318)	0.444 (1.321)
Bank capital	-0.024 (0.035)	-0.025 (0.035)	-0.024 (0.034)	0.459 (0.496)	0.552 (0.517)	0.561 (0.522)
Adj. R ²	0.133	0.116	0.111	0.125	0.132	0.130
Firm fixed effects	yes	yes	yes	yes	yes	yes
Number of Obs.	222,242	196,689	188,976	296,825	259,779	248,908

This table presents results of OLS regressions, where *New lending relationships* is a dummy that takes the value of one if the firm has a new loan as of 2009:Q2 with a bank that it had no loan as of 2007:Q2, and *Termination of existing lending relationship* takes the value of one if the firm had a loan with a bank as of 2007:Q2 and has no loan with the bank as of 2009:Q2. In Column 2, we interact interbank borrowing with a dummy for firms with total assets less than ten million euros, and the second interaction is that of interbank borrowing with a dummy for firms with total assets more than ten million euros (same for Column 5). In Column 3, we interact interbank borrowing with a dummy for firms with the number of employees less than one hundred, and the second interaction is that of interbank borrowing with dummy for firms with the number of employees more than one hundred (same for Column 6). Both total assets and number of employees are measured as of December 2006. *Interbank borrowing* is the ratio of interbank borrowing in 2007:Q2 for a bank. *Size* is the log of total assets of the bank. *Liquid assets* is the ratio of short-term to total assets of the bank. *Return on assets* is the profits of the bank divided by its total assets. *Nonperforming* loans is the fraction of loans that are in default as a fraction of total assets of the bank. *Capital* is the total bank capital as a fraction of total assets. Bank variables are as of 2007:Q2. The sample is restricted in all regressions to firms that have at least two banking relationships. Standard errors, reported in parentheses, are clustered at the bank level and are heteroscedasticity consistent. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

(a) Table 5: extensive margin, interbank borrowing, and firm size

Table 6
The extensive margin and strength of banking relationships

	Termination of existing lending relationship		
	(1)	Size (2)	No. of employees (3)
Interbank borr * dummy less than threshold		-0.055**	-0.052**
* relationship strength		(0.026)	(0.024)
Interbank borr * dummy more than threshold		-0.053***	-0.070***
* relationship strength		(0.014)	(0.018)
Interbank borr * relationship strength	-0.051**		
* relationship strength	(0.022)		
Interbank borrowing	0.519**	0.547**	0.524**
borrowing	(0.238)	(0.273)	(0.258)
Relationship strength	-0.066***	-0.070***	-0.069***
strength	(0.008)	(0.009)	(0.008)
Interbank borrowing * dummy more than threshold		0.120	0.308
borrowing * dummy more than threshold		(0.251)	(0.248)
Relationship strength * dummy more than threshold		0.041***	0.037***
strength * dummy more than threshold		(0.006)	(0.007)
Bank size	-0.021***	-0.022***	-0.021***
	(0.005)	(0.005)	(0.005)
Bank liquid assets	-0.043	-0.042	-0.044
	(0.086)	(0.086)	(0.087)
Bank return on assets	-3.293	-3.178	-3.423
	(5.365)	(5.308)	(5.387)
Bank nonperforming assets	0.334	0.399	0.368
	(0.935)	(0.934)	(0.935)
Bank capital	0.042	0.027	0.052
	(0.364)	(0.366)	(0.369)
Adj. R ²	0.232	0.234	0.234
Firm fixed effects	yes	yes	yes
Number of obs.	259,779	259,779	248,904

This table presents results of OLS regressions, where *Termination of existing lending relationship* takes the value of one if the firm had a loan with a bank as of 2007:Q2 and has no loan with the bank as of 2009:Q2. In Columns 2 and 3 we interact with interbank borrowing and relationship strength with a dummy for firms with total assets less than ten million euros, and the second interaction is with a dummy for firms with total assets more than ten million euros. In Columns 3 and 4, we interact interbank borrowing and relationship strength with a dummy for firms with the number of employees less than one hundred, and the second interaction is with a dummy for firms with the number of employees more than one hundred. Both total assets and number of employees are measured as of December 2006. Relationship strength is the log of credit outstanding between a firm and a bank as of 2007:Q2. *Interbank borrowing* is the ratio of interbank borrowing in 2007:Q2 for a bank. *Size* is the log of total assets of the bank. *Liquid assets* is the ratio of short-term to total assets of the bank. *Return on assets* is the profits of the bank divided by its total assets. *Nonperforming* loans is the fraction of loans that are in default as a fraction of total assets of the bank. *Capital* is the total bank capital as a fraction of total assets. Bank variables are as of 2007:Q2. The sample is restricted in all regressions to firms that have at least two banking relationships. Standard errors, reported in parentheses, are clustered at the bank level and are heteroscedasticity consistent. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

(b) Table 6: extensive margin and strength of banking relationships

6.5 Tables 7 and 8: bank fundamentals, central bank borrowing, and liquidity hoarding

Read:

- Table 7 asks whether the credit-supply effect varies by bank characteristics.
- The interaction between bank nonperforming loans and interbank borrowing is negative, meaning the liquidity shock is more damaging at less solvent banks.
- The interaction with bank capital is also negative; the paper interprets this cautiously because riskier banks may hold more capital due to regulation or market pressure.
- Central bank borrowing is positively associated with credit supply, but the interaction with interbank borrowing is negative. Liquidity support does not fully offset the shock for banks most dependent on interbank funding.
- Table 8 shows that interbank borrowing does not significantly reduce total bank assets, but it is positively associated with changes in bank liquid assets.
- This pattern suggests that exposed banks cut firm credit and increase liquidity buffers rather than shrinking total assets.

Economic message: Bank liquidity and bank solvency interact. Central bank liquidity may be hoarded by the most exposed banks rather than fully passed through to firms.

Table 7
Differential effects of interbank dependence on credit supply by bank characteristics
Change in the log level of firm-bank total credit volume between 2009:Q2 and 2007:Q2

	(1)	(2)	(3)	(4)	(5)	(6)
Interbank borrowing	-0.832 (1.957)	-0.442*** (0.155)	-0.149 (0.172)	-0.292** (0.124)	0.034 (0.203)	-0.373*** (0.132)
Bank size * interbank borr	0.021 (0.089)					
Bank liq assets * interbank borr		0.616 (0.918)				
Bank ROA * interbank borr			-47.013 (34.247)			
Bank NPL * interbank borr				-18.615* (9.582)		
Bank capital * interbank borr					-6.304** (2.694)	
Central Bank borr						0.027*** (0.009)
Central bank borr * interbank borr						-0.081** (0.033)
Adj. R^2	0.095	0.095	0.096	0.096	0.097	0.114
Firm fixed effects	yes	yes	yes	yes	yes	yes
Number of obs.	220,284	220,284	220,284	220,284	220,284	182,343

This table presents results of OLS regressions, where the dependent variable is the change in the (log) level of committed credit volume for each firm-bank pair between 2009:Q2 and 2007:Q2, where the initial interbank shock was August 2007. *Interbank borrowing* is the ratio of interbank borrowing to total assets in 2007:Q2 for a bank. *Size* is the log of total assets of the bank. *Liquid assets* is the ratio of short-term to total assets of the bank. *Return on assets* is the profits of the bank divided by its total assets. *Nonperforming* loans is the fraction of loans that are in default as a fraction of total assets of the bank. *Capital* is the total bank capital as a fraction of total assets. Total firm-bank credit is the log of committed credit outstanding between a firm and a bank as of 2007:Q2. *Central bank borr* is the liquidity availed by a bank from the European Central Bank. Bank variables are as of 2007:Q2. All regressions include controls for bank characteristics (unreported). Firm-level variables are measured as of December 2006. Standard errors, reported in parentheses, are clustered at the bank level and are heteroscedasticity consistent. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

(a) Table 7: differential effects by bank characteristics

Table 8
Rebalancing of bank portfolio

	Change in bank total assets (1)	Change in bank liquid assets (2)
Interbank borrowing	0.128 (0.186)	0.066* (0.033)
Bank size		0.001 (0.005)
Bank liquid assets	-0.171 (0.323)	
Bank return on assets	-18.369 (12.469)	3.718* (2.117)
Bank nonperforming assets	0.020 (1.326)	0.263 (0.363)
Bank capital	0.700 (0.756)	0.173 (0.149)
Adj. R^2	0.155	0.107
Number of obs.	80	80

This table presents results of OLS regressions, where the dependent variable is the change in the level of bank total assets and bank liquid assets between 2009:Q2 and 2007:Q2, where the initial interbank shock was August 2007. *Interbank borrowing* is the ratio of interbank borrowing to total assets in 2007:Q2 for a bank. *Size* is the log of total assets of the bank. *Liquid assets* is the ratio of short-term to total assets of the bank. *Return on assets* is the profits of the bank divided by its total assets. *Nonperforming* loans is the fraction of loans that are in default as a fraction of total assets of the bank. *Capital* is the total bank capital as a fraction of total assets. Bank variables are as of 2007:Q2. Robust standard errors, reported in parentheses, are heteroscedasticity consistent. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

(b) Table 8: rebalancing of bank portfolio

6.6 Table 9: substitution and binding credit constraints

Read:

- Table 9 moves from firm-bank pairs to the firm level.
- The key variable is the firm's average exposure to banks with high interbank borrowing, weighted by the precrisis credit share of each bank.
- Firms borrowing more from exposed banks experience larger reductions in total bank debt.
- The same exposure also predicts lower total firm debt in 2008, even after controlling for prior debt and firm characteristics.
- Results are similar using average foreign interbank borrowing, consistent with the idea that foreign interbank funding dried up especially strongly.

Economic message: Firms cannot fully replace credit lost from affected banks. The bank liquidity shock translates into a binding firm credit crunch, not merely a reshuffling of loans across banks.

Table 9
Credit substitution, total supply reduction, and interbank borrowing

	Change in total bank debt (1)	Change in total bank debt (2)	Total firm debt, 2008 (3)	Total firm debt, 2008 (4)	Change in total bank debt (5)	Total firm debt, 2008 (6)
Average interbank borrowing	−0.576*** (0.032)	−0.636*** (0.043)	−0.510*** (0.032)	−0.037*** (0.018)		
Average foreign interbank borr					−0.361*** (0.052)	−0.074*** (0.023)
Firm ROA		−0.189*** (0.065)		−0.003 (0.012)	−0.145** (0.065)	−0.003 (0.012)
Firm age		−0.056*** (0.009)		−0.076*** (0.003)	−0.046*** (0.009)	−0.075 (0.003)
Firm size		−0.093 (0.253)		0.164*** (0.039)	−0.089 (0.253)	0.164*** (0.039)
Total firm debt				0.926*** (0.003)		0.925*** (0.023)
Adj. R^2	0.008	0.011	0.365	0.845	0.009	0.845
Firm-industry fixed effects	yes	yes	yes	yes	yes	yes
Firm-location fixed effects	yes	yes	yes	yes	yes	yes
Number of obs.	108,878	74,912	70,958	61,312	74,912	61,312

This table presents results of OLS regressions, where the dependent variable in Columns 1, 2, and 5 is the change in the log of total bank loan volume for each firm between 2009:Q2 and 2007:Q2 (where the initial interbank shock was in August 2007), whereas in Columns 3, 4, and 6 it is the log of overall debt of the firm outstanding as of December 2008 (includes bank debt and other forms of debt from suppliers as trade credit, etc.). *Average interbank borrowing* is, for each firm, the weighted (by loan size) average level of interbank borrowing of the banks in May 2007 that are lending to the firm in 2007:Q2. *Average foreign interbank borrowing* is the same as *Average interbank borrowing* but only using the interbank funds borrowed from foreign banks. *Firm ROA* is the return of assets of the firm. *Firm age* is the age of the firm measured using the date of the firm's incorporation ($\ln(1+\text{age})$). *Total firm debt* is the log of total firm debt outstanding (including all forms of debt). *Firm size* is the log of assets of the firm. Firm-level controls are measured as of December 2006. All regressions are estimated with dummies for geographic location and the industry of the firm. The sample is restricted in all regressions to firms that have at least two banking relationships. White heteroscedasticity consistent robust standard errors are reported in parentheses. ***, **, * indicate significance levels at 1%, 5%, and 10% respectively.

Table 9: credit substitution, total supply reduction, and interbank borrowing

7 Short summary

- The paper studies the 2007–2009 crisis using exhaustive Portuguese loan-level data.
- Banks more reliant on interbank borrowing before the crisis cut credit more after the interbank market froze.
- Firm fixed effects compare different banks lending to the same firm, helping isolate bank credit supply from firm credit demand.
- A 10 percentage point increase in precrisis interbank borrowing is associated with about a 3.7–4 percent additional credit reduction.
- The credit contraction is concentrated among small, young firms and firms with weaker banking relationships.
- Liquidity-exposed banks are less likely to form new lending relationships with small firms.
- Strong bank-firm relationships reduce the likelihood that an existing lending relationship is terminated.
- Banks with weaker fundamentals, especially higher nonperforming loans, transmit the liquidity shock more strongly.
- Central bank liquidity has limited evidence of pass-through to exposed banks' firm credit supply and is associated with liquidity hoarding.

- Firm-level regressions show that firms cannot fully substitute lost bank credit with other bank credit or nonbank debt.
- The main contribution is to show a causal transmission channel from interbank liquidity stress to a binding credit crunch for entrepreneurial firms.

8 Conclusion

8.1 Core conclusion

Iyer, Peydró, da-Rocha-Lopes, and Schoar show that the 2007–2009 interbank liquidity crunch reduced firm credit supply through exposed banks. Banks with higher precrisis interbank borrowing cut lending more during the crisis. The effect is not merely an aggregate decline in loan demand: it remains when comparing different banks lending to the same firm.

8.2 Economic intuition

Banks that relied on interbank funding faced a sudden funding shortage when the interbank market froze. To conserve liquidity, they reduced credit supply to firms. Small and young firms were hit hardest because they had fewer alternative financing options and weaker relationships. Larger firms were better protected, either because they were more valuable to banks or because they could access alternative sources of credit.

8.3 Incremental contribution revisited

The paper’s contribution is the complete micro-level mapping of the crisis transmission mechanism. It does not simply show that bank credit fell during the crisis. It shows that banks hit harder by an external liquidity shock cut credit more, that the effect is identified within the same borrowing firm, that entrepreneurial firms were most exposed, and that the shock was binding because firms could not fully substitute other debt.

8.4 Policy implications

The results caution that broad central bank liquidity provision may not automatically translate into credit for the firms most in need. If banks are worried about solvency or future liquidity, they may hoard liquidity rather than lend. Policies aimed at supporting entrepreneurial firms may need stronger mechanisms that directly encourage or require credit pass-through to small and medium enterprises.

8.5 Limitations

The paper studies Portugal and a particular crisis, so the strength of the channel may vary across banking systems. The identification is strong for relative bank credit supply but less direct for

general equilibrium effects. The central-bank liquidity analysis is not experimental. Finally, committed credit may differ from actual credit usage, especially when firms draw down lines during crises.

8.6 Takeaway

The enduring takeaway is that wholesale funding fragility can become a real-economy credit constraint. When interbank liquidity dries up, exposed banks cut credit; when credit is relationship-based and small firms lack substitutes, the financial shock becomes a firm credit crunch.